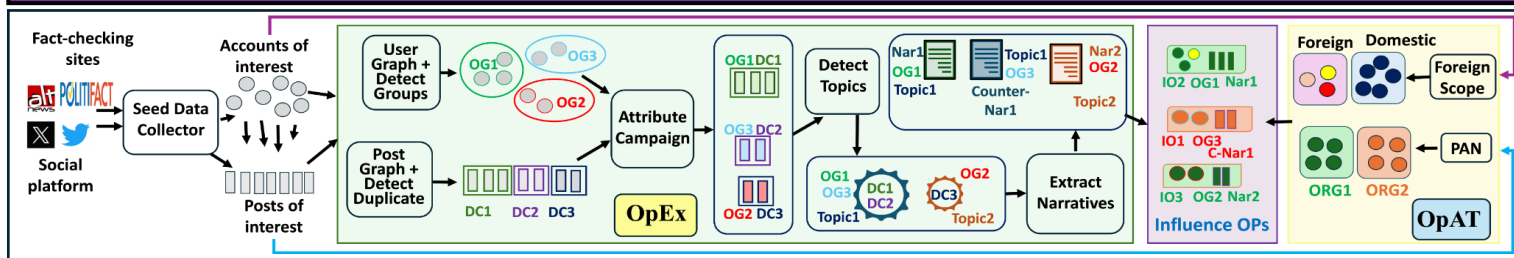


PROBLEM

- Independent IO oversight limited by restricted platform access
- Researchers lack firehose-scale visibility into platform activities and evolving campaigns
- Platform IO disclosures are selective, retrospective, and opaque about detection methods
- Can public data support transparent, reproducible detection and attribution of influence operations?**

IO-X PIPELINE



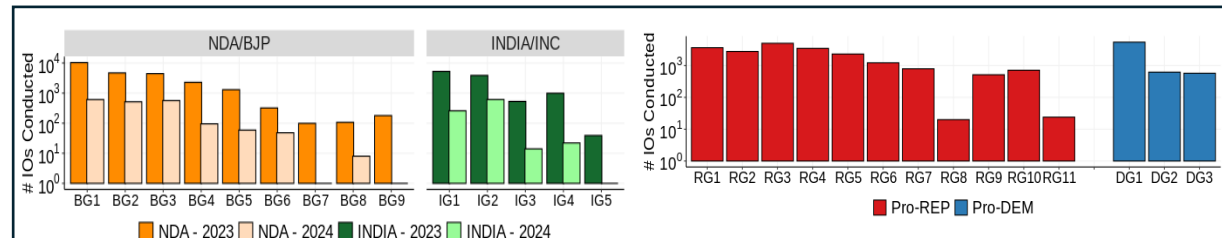
KEY IDEA

- Search social media **X/Twitter** for accounts that duplicate fact-checker verified false claims
- Monitor accounts longitudinally
- Find account communities that amplify the same content

DETECTION AT SCALE

- IO-X detects substantially more duplication campaigns than prior methods
- US Elections'24: 74,598 campaigns
31.5× / 18.7× vs. Pacheco / Vishnuprasad
- INDIA'23: 198,457 campaigns
43.9× / 10.3× vs. Pacheco / Vishnuprasad

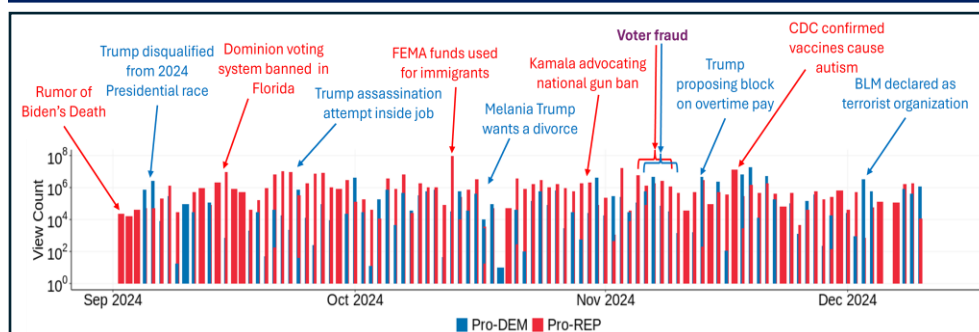
PARTISAN IO STRUCTURE



CONTRIBUTIONS

- Formal model:** IO detection = undetectability game
- Theory:** content duplication creates gap between coordinated operations and background activity
- System:** IO-X uses public data to monitor, identify and attribute IOs
 - PAN** infers political alignment
 - ForeignScope** detects foreign-linked accounts
- Main Takeaway:** IO-X detected 64k+ IOs across U.S. and Indian election datasets

FALSE CAMPAIGNS & REACH



WHY IT MATTERS

- IO-X identified **957** false-information campaigns that were part of IOs
- 691** promoted by Pro-REP groups; **266** by Pro-DEM groups
- Pro-REP campaigns received **324M** views; Pro-DEM campaigns received **83M** views